

Depth, Salinity and Temperature Measurements at Dana Point

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Abstract— This paper will describe the design, construction and experimental results using an autonomous underwater vehicle (AUV) for Harvey Mudd College's Experimental Engineering Course (E80). The AUV measured temperature, salinity, and depth at various locations at Dana Point Harbor in Southern California. Data was collected using a Teensy microcontroller and analyzed using MATLAB. Temperature was measured using a thermistor sensor and steadily decreased from 21.25 to 20.6 ±0.22°C as the robot moved further from the shore. Salinity was measured using a Nema 1-15p wall plug, and measurements were similar in offshore and nearshore deployments and constant at 33.4 ±0.2ppt. Depth was measured using sonar time of flight method and measured at nearshore depth of 3.93 ft ± 0.7 ft. It was concluded that temperatures and salinity could be measured with reasonable accuracy while depth could only be measured with reasonable accuracy in shallower depths with less complex floor structures. These results demonstrate that while low-cost salinity and temperature sensors can be accurate for ocean measurements, acoustic depth sounding requires signal processing to achieve accuracy complex seabed environments.

I. INTRODUCTION

Measurements such as temperature and salinity can be used to monitor environmental changes associated with climate change and coastal runoff. Differences in temperature and salinity impact water density, which affects how well ocean water mixes. Salinity levels that indicate reduced mixing can limit how much oxygen is able to reach deeper waters which can create dead zones that can destroy ecosystems [1]. Salinity measurements can also detect freshwater runoff and indicate areas that might be negatively influenced by human activity [2]. Warmer temperature measurements can be used as an indicator of climate change. Furthermore, depth measurements are essential for the field of bathymetry and are used for navigation and exploration [3].

Temperature was measured using a thermistor, a resistor whose resistance changes with temperature. Salinity was measured using a conductivity probe as conductivity is a proxy for ocean salinity. Depth was measured using sounding a method where a wave pulse is sent using a speaker toward the ocean floor and a microphone picks up an echo. Then using the speed of sound in water, and the time between the pulse and the echo, the depth can be calculated [4].

The main goals of this project were to:

- Design and integrate a group of sensors capable of measuring environmental factors in the ocean
- Compare nearshore and offshore water measurements to identify environmental differences.
- Validate the accuracy of low-cost custom sensors against established ground truths.

II. SENSOR DESIGN

A. Depth

A sonar time of flight method was used to measure the depth. In this method, a speaker sends out a short pulse which travels toward the ocean floor. The pulse bounces off the ocean floor and is recorded by the microphone. The depth is calculated using the following equation:

$$Depth = V \frac{\Delta T}{2} \quad (1)$$

Where V is the speed of sound, assumed to be 1500 m/s [5] in seawater and ΔT is the time in between when the pulse is sent out and when it is heard by the microphone. The depth at Baby Beach was estimated to have a range of 0-5 ft near shore and 9-15 ft in the harbor [6].

A DAEX25W-8 speaker was used to send a pulse and a CME-1538-100LB electret microphone was used to receive the pulse. They were chosen due to easy access, cost and waterproof characteristics. The speaker circuit is shown below in Figure 1.

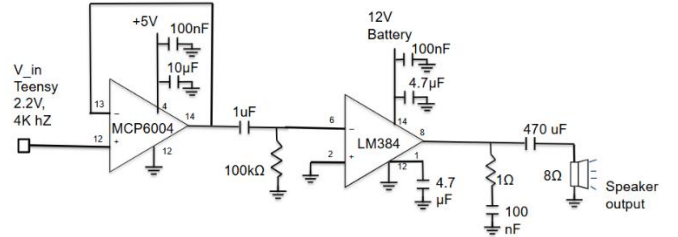


Figure 1. Speaker Circuit Schematic

The goal of this circuit is to send out a pulse that will bounce off the ocean floor and allow the depth to be measured. The circuit's input is a 1ms pulse of a ~2.2Vpp, 4kHz square wave sent by the Teensy microcontroller. The pulse is then sent through a unity gain buffer which was used to provide high input impedance to prevent loading from the Teensy output. This output is then sent through a passive high pass filter which changes the signal to be more like a sine wave by cutting off low frequencies of the square wave. This signal is then sent to an LM384 Audio amplifier powered directly by a 12V battery. This amplifier has a voltage gain fixed at 50 with the goal of making the signal as loud as possible. The output from the amplifier is sent to a DC bias removal using another high pass filter, centering the signal around zero. The output is sent to the DAEX25W-8 speaker where the tone is played. This tone and its echo are recorded by the microphone, whose circuit is shown below in Figure 2:

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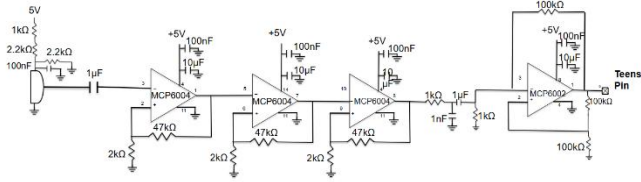


Figure 2. Microphone Circuit Schematic.

The goal of this circuit is to pick up the signal sent by the speaker and amplify it enough to be read. The signal is recorded by the microphone on the left. The signals are captured in the order of millivolts so they must be amplified significantly to pick up reasonable measurements. Therefore, the signal is sent through three gain stages using non-inverting amplifiers. Each amplifier has a gain of 24 giving a total gain of around 13,824. Signal clipping is not a concern for this circuit as the goal of the circuit is only to pick up a pulse, not measure its voltage. The signal is then sent through a filter which filters out frequencies greater than 6.8kHz. This ensures the microphone doesn't pick up any other noise in the area such as boats and wildlife. Finally, the output is sent to a Teensy analog pin.

Reading these pulses requires reading frequency much faster than the Teensy analog pin default of around 10Hz. Therefore, BurstADCSampler code was used to temporarily increase the sampling rate to around 200kHz. This code is run right after the speaker pulse is sent out and samples data at a higher frequency for short bursts allowing the Teensy to pick up the echo from the speaker. This is then saved as a separate file and can be read using Excel.

This sensor was calibrated through tests in the tank room and Axelrood Pool. An oscilloscope trace from a test in the tank room is shown below in Figure 3:



Figure 3. Oscilloscope trace of tank room testing showing speaker output (red) and mic reading (green)

The green spikes indicate the pulse reflecting off the tank floor and ocean surface. The time between the two peaks is chosen to calculate the depth using Equation 1. The trace shows a difference of about 1.8ms between mic input spikes indicating a depth of around 1.35m in the tank. This assumes the speed of sound in the tank is the same as it is in the ocean at 1500m/s [5]. This depth measurement of 1.35 is consistent with depth of the tank, validating the method.

B. Salinity Sensor

One method to measure the salinity, or salt content, of ocean water is to measure the conductivity of the water. With higher conductivity being proportional to higher salinity. Conductivity is inversely related to resistance, so measuring the 'resistance' between two conductive plates in the ocean allows for a calculation of salinity. According to NOAA, The expected range of salinity in the ocean is 33-37ppt (parts per thousand) [7].

A NEMA 1-15p wall plug was used as a probe to measure the salinity. It was chosen for its availability and ease of integration. The prongs of the connector were removed in order to improve accuracy of measurements. The schematic for the circuit is shown below in Figure 4:

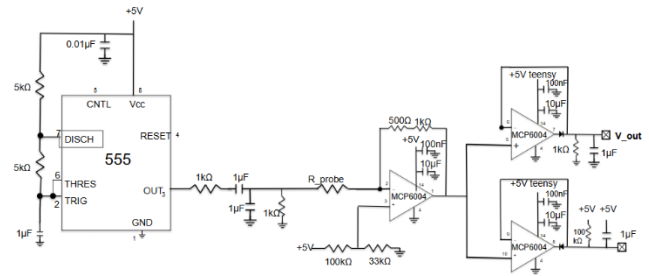


Figure 4. Salinity sensor schematic where R_{probe} is the NEMA 1-15p wall plug

The main goal of the circuit is drive an AC signal through the probe, and depending on the resistance of the probe the gain from the non-inverting offset amplifier will change. Finally, the amplitude of the amplified AC signal measured using max and min detectors. One of the largest problems in measuring salinity is corrosion. In order to combat this, a 555 timer drives AC signal through the probe. The 555 is powered by 5V and drives a 200Hz, 50mV square wave. The signal is then filtered by a high and low pass filter in order to make the signal more sinusoidal. The signal is then sent through the probe where the salinity is measured as a resistance. This resistance acts as a variable resistance depending on the salinity. Since the gain of amplifier is dependent on the resistance of the probe and resistance of the probe is unknown, trial and error was conducted with different feedback resistor values until a suitable gain was achieved for expected salinity ranges. Furthermore, the offset of 1.24V was chosen to center voltages within the range readable by the Teensy (0-3.3V). Finally, after the amplifier stage, the max and min detectors hold the maximum and minimum values of the wave signal. This allows the Teensy to read the amplitude of the signal. The max and min values are read by the Teensy and the difference between them is related to the salinity using the calibration curve in Figure 5:

Salinity Calibration Curve

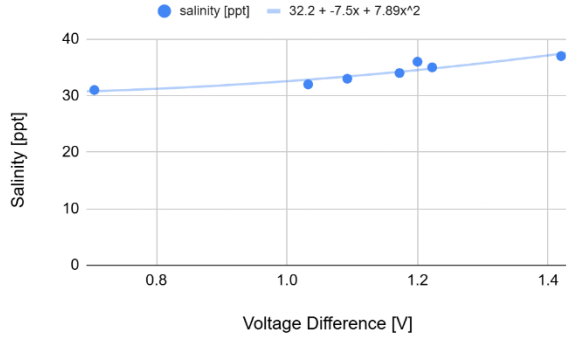


Figure 5. Calibration Curve showing how voltage difference is related to measured salinity.

The calibration curve was created by preparing solutions of known salinities from 30-37ppt. The solutions were created by measuring the mass of the water and salt added to the solutions. The voltage difference at each salinity value was measured and a curve was fit using a polynomial fit.

The maximum and minimum values were read on A02 and A03 respectively. Their difference was calculated and converted to salinity using the calibration equation in Equation 2.

$$\text{Salinity} = 32.2 - 7.5 * D + 7.89 * D^2 \quad (2)$$

Where D represents the difference in voltage read by A02 and A03. Since the salinity is not expected to fluctuate on a millisecond scale, the default 10 Hz sample rate was sufficient.

C. Temperature Sensor

Temperature was measured using a Murata NXFT15WB473FA2B150 47k thermistor. This thermistor was chosen due to its accuracy of up to $\pm 0.1^\circ\text{C}$ and relatively fast response time when compared to other temperature sensors such as linear sensors. Furthermore, thermistors are much easier to integrate than a thermocouple as they do not require a reference temperature for calibration [8]. The fast response time and high accuracy are beneficial when difference in ocean temperatures is small. The expected range of ocean temperatures at Dana Point are $15\text{-}21^\circ\text{C}$ [9]. The Schematic for the sensor is shown below in Figure 6:

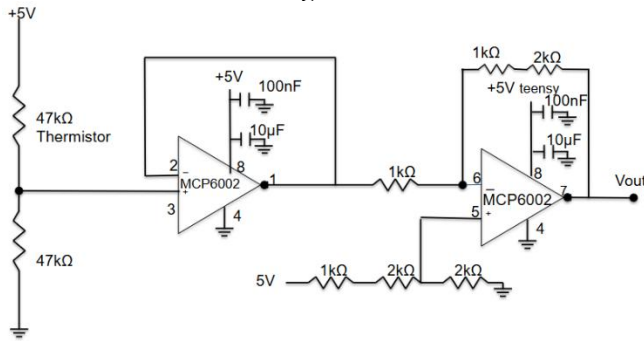


Figure 6. Thermistor circuit schematic

The schematic shows the thermistor in a voltage divider sent to a unity gain buffer then to a inverting offset amplifier. The unity gain buffer was used to provide a high input impedance to prevent loading the thermistor voltage divider, ensuring the measured voltage remained a true representation of the resistance change.

Since a thermistor is a resistor whose resistance changes with temperature and the Teensy's analog pins can only read voltages, a voltage divider was created to relate the resistance of the thermistor into a readable voltage. The Schematic and equation for a voltage divider is shown below in Figure 7:

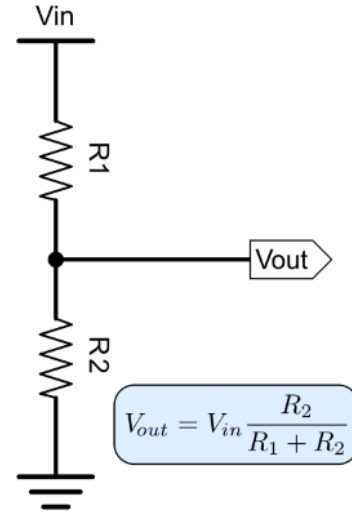


Figure 7: Voltage divider schematic and equation where R1 is the thermistor, and R2 is a constant. Retrieved from [10]

Assuming a constant V_{in} of 5V, the output of the voltage divider can be determined using an inverse relationship of with the thermistor's resistance. The output of the voltage divider is then sent into a unity gain buffer whose output sent into an inverting offset amplifier. The goal of the offset amplifier is to make the output voltage range as close to 0.3-3V as possible to ensure the Teensy is able to read all temperature values. The resistance values for the op amp were chosen using Equation 3 shown below:

$$V_{out} = \left(1 + \frac{R_f}{R_{n1}}\right) \left(\frac{R_g}{R_{p1} * R_g}\right) * V_{p1} - \frac{R_f}{R_{n1}} * V_{n1} \quad (3)$$

Where the locations of resistors and voltages are shown below in Figure 8:

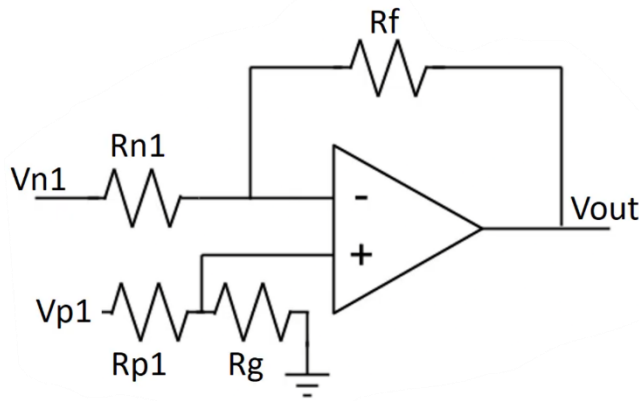


Figure 8. Schematic showing locations of resistors and voltages referenced in Equation 3. Retrieved from [11]

In order to select resistance values for the inverting offset op amp, a temperature range of 10-25 °C was chosen to reduce risk of measurements railing out. According to the thermistor's data sheet [12], the expected resistance range given a temperature range of 10-25°C is 47-80k. This is the range of R1 shown in Figure 7. The output range of Figure 7's voltage divider is then 1.8-2.5V. The selected resistors in Figure 6, scale shift and scale the 1.8-2.5V input range to 0.5-3.08V output range which is read by the Teensy A10 pin.

In order to convert voltage measurements into a temperature a calibration curve was created. The use of a Steinhart-Hart equation was attempted, but due to issues with coefficient calibration a simple calibration curve method was chosen. Solutions of different temperatures from 11.7-22.4°C were created and measured using a thermometer. The output voltage was measured using the were used to create the calibration curve. The curve and equation are shown below in Figure 9:

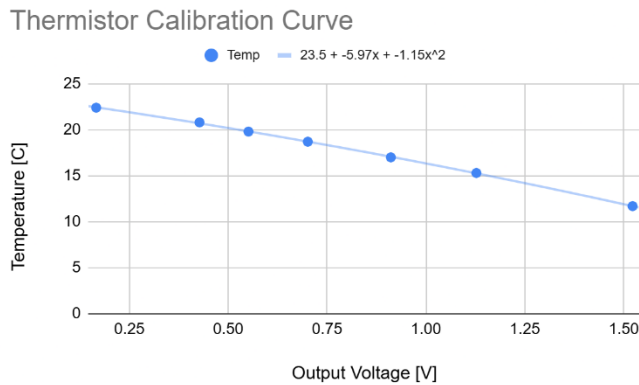


Figure 9. Calibration curve relating voltage to temperature.

The voltage measurements read by Teensy A10 pin and then converted to a temperature measurement using the equation in Figure 9. Since the temperature is not expected to fluctuate on a millisecond scale, the default 10 Hz sample rate was thought to be sufficient.

III. ROBOT DESIGN

The AUV was designed to be a robot which floats on the surface and can move in one direction for a set amount of time. An image of the robot is shown below in Figure 10:

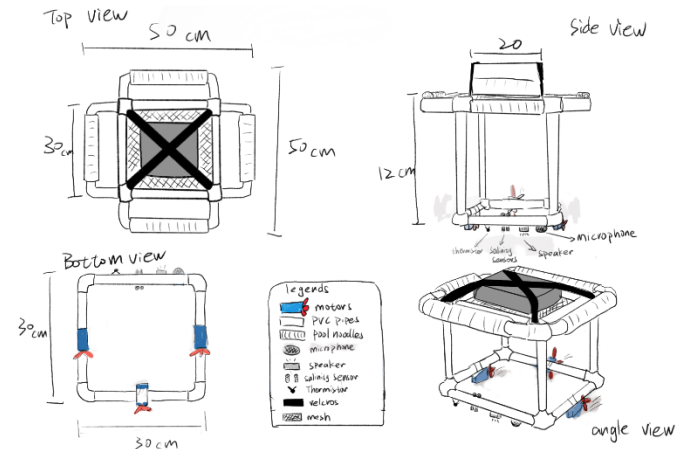


Figure 10. Mechanical drawing of robot design

One of the main objectives of the robot design was to reduce the risk of water getting into the circuitry. As such, the waterproof box was placed on top of the robot and sat above the surface at all times. Plugs were drilled into the box allowing wires to pass in and out of the box and plugs were waterproofed using epoxy. This box was secured using velcro to reduce risk of box sliding off the top of the robot. The box contained the motherboard, battery and protoboard with all circuitry and is shown in Figure 11:



Figure 11. Picture showing waterproof box

The sensors and motors were placed 12cm below the surface to ensure they would always be under the water even in turbulent conditions. Their layout is shown below in Figure 12:

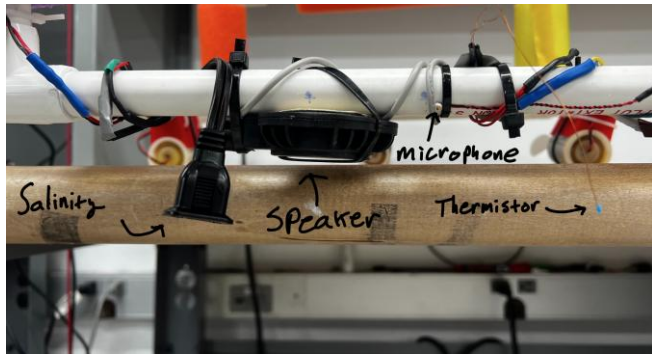


Figure 12. Close up view showing sensor placement

The speaker and microphone were placed close to each other to ensure the microphone could pick up the initial pulse sent out by the speaker. They were both placed facing straight down to ensure the sound was bouncing off the ocean floor. The salinity probe and thermistors were placed on the outer edges to allow for easy replacement during operation.

Since the robot stayed on the surface, there was no need for buoyancy calculations. The goal was to ensure the top frame of the robot would stay at the water level and the sensors would sit below the water level. Therefore, pool noodles were placed around the frame of the top frame as seen in Figure 10. This set up was tested in the tank room to ensure the box would sit above the surface.

Three motors were attached to the frame with one on each side and one in the middle. This ensured the robot moved straight and was strong enough to fight currents or small waves. The motors were attached to the frame using custom 3D printed parts and zip ties. The CAD files for the 3D printed parts are shown below in Figure 12:

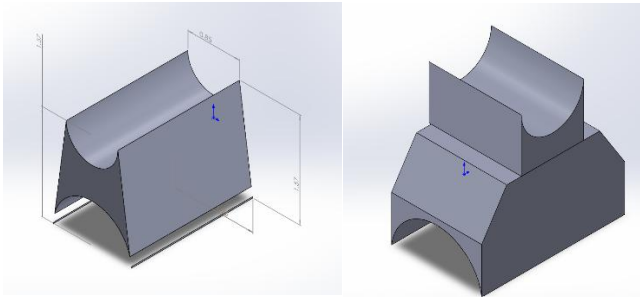


Figure 13. CAD file showing part connecting motor to side of frame (left) and back of frame (right)

These mounts allowed the motor to be mounted directly on the frame and to stay in place during deployment.

IV. DEPLOYMENT DETAILS

The AUV was deployed and tested at pHake Lake, Axelrood Pool and Dana Point. For all deployments, the following experimental procedure was conducted. The code was uploaded to the Teensy from a computer, the SD card was inserted into the motherboard, the battery was turned on, the water proof box was closed and fastened with velcro and the robot was placed in the water. Once the battery was turned on, the robot sat still 60s before moving allowing time for users to prepare for data

collection. After data was collected, the robot played a loud noise on the speaker signal it had finished and it was pulled back using the leash visible in Figure 14, operators dried their hands and opened the water proof box and turned off the battery. The salinity sensor was wiped off using a paper towel. The SD card was then removed and data was loaded into MATLAB for analyzation.

The calibration curves in Figures 5 and 9 and historical data were used to estimate the voltage output range. This ensured the damaging voltages weren't sent to the Teensy. The robot was chosen to collect data for 60s because this would be enough time for all sensors to sample data and reach a steady state value.

The first deployment was at pHake Lake located in the Bernard Field Station (BFS) in Claremont, California. The robot was taken out to the middle of the lake using a boat and deployed in one direction. The goal for this deployment was to test the GPS integration of the robot. However, due to GPS data being too noisy to work with, a simple timer was chosen. For simplification, the robot moved forward 30s, stopped, took data for 60s and repeated two more times. This allowed three sets of data at different locations for testing.

The next deployment was conducted at Axelrood Pool at Claremont McKenna College in Claremont, California. The goal of this deployment was to test the depth sensor at greater depths than the tank room as well as with more background noise. The AUV was tested at both the shallow end (7ft) and deep end (15ft). The shallow end test is shown in Figure 14 below:

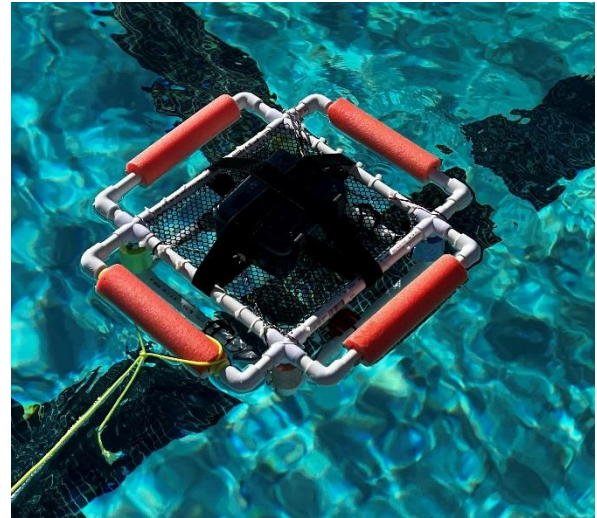


Figure 14. Deployment in the Axelrood Pool

After the deploying, the robot the microphone's datalog files were analyzed and the depth was calculated using Equation 1 in Section II. The results of the shallow end test are shown below in Figure 15:

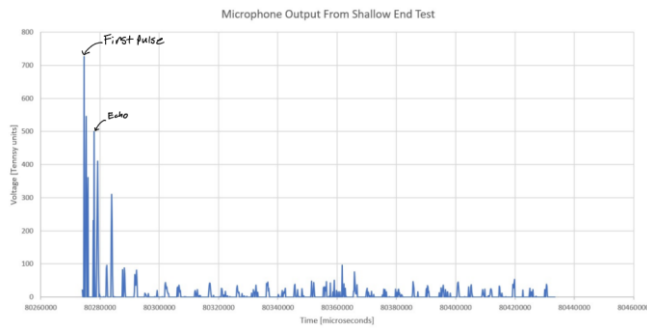


Figure 15. Microphone output from testing in Axelrood Pool's shallow end

The test shows the first pulse at $80274655\mu\text{s}$ and an echo heard at $80278005\mu\text{s}$ giving a ΔT of 0.00335s . Using Equation 1, a depth of 2.5125m (8.24ft) was calculated giving an error of 1.24ft or 17.7% assuming the depth in the shallow end was 7ft . The depth in the deep end was also tested and using the same procedure explained above, the depth was measured at 20.8ft giving an error of 5.8ft or 38.7% . The main takeaway from this test is that the depth sensor displayed an overall error range of $17.7\% - 38.7\%$ from shallow to deep range but was still able to measure the relative changes in depth.

The final deployment was conducted at Baby Beach in Dana Point, California. The AUV was deployed at multiple locations including right off the beach and closer to the middle of the harbor using a kayak. The deployment locations are shown below in Figure 16:



Figure 16. Screenshot from Google Earth showing locations of deployment marked by stars

For all deployments, the robot traveled out to sea (south) and was then retrieved using the leash attached seen in Figure 14. After deployment, the SD card was retrieved and data was analyzed using MATLAB and Excel. The main goal of this test was to compare nearshore measurements to deeper

measurements. An image of a near shore deployment is shown below in Figure 17.



Figure 17. Nearshore AUV deployment at Dana Point

Personal safety was ensured by all team members wearing life vests for kayak deployments. Robot safety was ensured by checking the waterproof box was tightly sealed and velcroed on before deployment and attaching a leash to the robot. Furthermore, the AUV was deployed far way from people paddling or playing in the water and the robot was followed by a team member in a kayak.

V. RESULTS

The salinity sensor was the most consistent of the data and data was collected in both nearshore and offshore deployments at Dana Point. A graph of both nearshore and offshore salinity measurements are shown below in Figure 18.

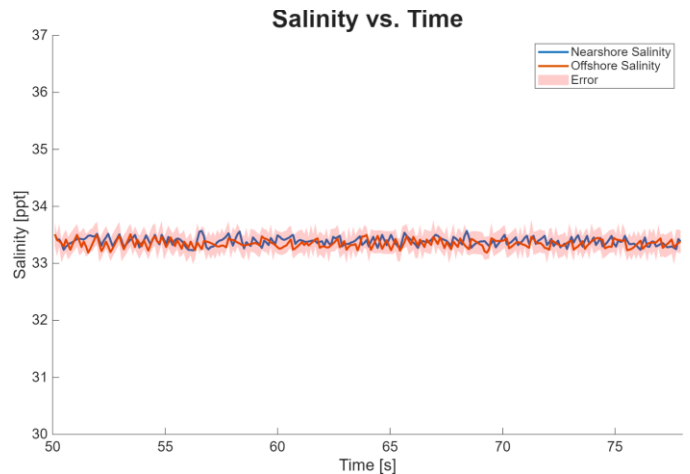


Figure 18. Salinity vs. time measurements show overlapping nearshore and offshore tests

The results show a near constant salinity around 33.4ppt for both nearshore and offshore tests. The error of $\pm 0.2\text{ppt}$ represents the electrical noise of the measurements observed during testing. These measurements fit within the expected range of ocean salinities of $33\text{-}37\text{ppt}$ [7]. For both tests, the robot stayed still and collected data over time. The relatively stable

salinity measurements indicates a well circulated system with low amounts of river runoff. River runoff tends to decrease salinity nearshore as the river's freshwater has a lower salinity than the ocean water, so a similar salinity between nearshore and offshore tests indicate a lack of runoff in the tested locations.

The temperature sensor worked only for earlier tests located near the shore. It was hypothesized that this was due to salt water seeping into the wiring and corroding the sensor. Figure 19 below shows the temperature vs. time for a nearshore run. Since the AUV started closer to shore and moved farther out, the time relates to distance from shore.

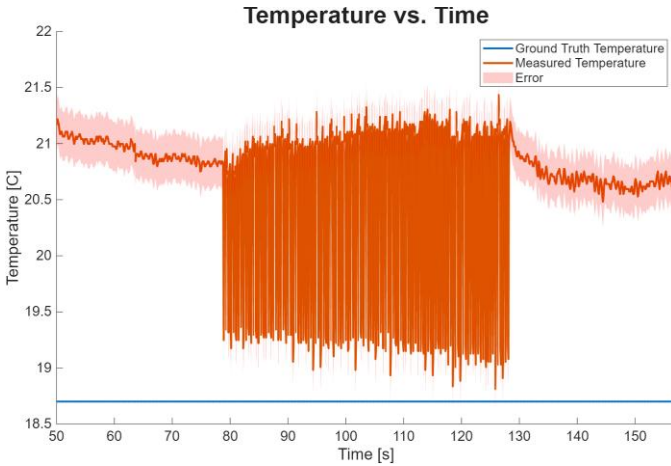


Figure 19. Temperature vs. Time measurements for nearshore test

The results show the temperature decreasing over time from around 21.25 to 20.6 °C. The error was calculated by assuming $\pm 0.2^\circ\text{C}$ electrical noise as well as the sensors inherent accuracy of $\pm 0.1^\circ\text{C}$, giving a total error of $\pm 0.22^\circ\text{C}$. The noisy data from around 80 to 130s was when the motor was running which causes voltage measurements to be noisy. The temperature measurements were higher than the ground truth measurement probed at 18.7°C near the pier. Some reasons for discrepancy between the measured data and the ground truth measurement could be the ground truth measurement being taken at the pier, farther away from the shore than where the AUV was deployed and inconsistency with the calibration curve as the curve was created using a different thermistor than the one used to collect data. However, the trend of decreasing temperature over time shows that the temperature decreases as the robot moves further away from the shore. This lines up with the hypothesis that ocean temperatures will decrease as the farther away from the shore due to the heat capacity of water. However, the errors are overlapping for all tests making it hard to draw any conclusions about the relationship of distance from shore and temperature.

Depth was tested at the nearshore and offshore locations shown in Figure 16. The results of a nearshore test are shown below in Figure 20:

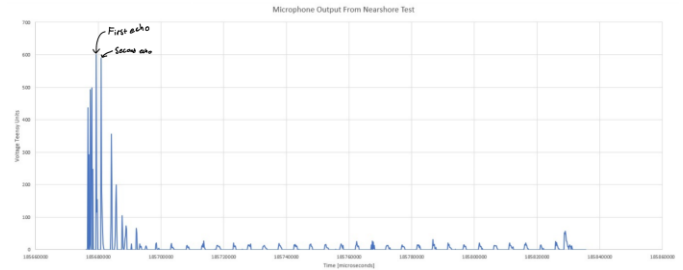


Figure 20: Microphone output from nearshore test

The data had a lot of noise and it was hard to tell which peaks to choose. The two highest amplitude peaks were chosen to analyze. Their times were $185679368\mu\text{s}$ and $185680965\mu\text{s}$ giving a ΔT of 0.001597s . Using Equation 1, a depth of 1.20m (3.93ft) was calculated. Using error calculations from Section V, the uncertainty of this measurement is $\pm 0.7\text{ft}$ assuming the 17.7% error in shallow measurements. This measurement made some sense as the water was about waist high during data collection. However, due to the many peaks of the data, it is hard to tell which points are echoes from pulses sent out from speaker and background noise picked up by the microphone. The results of the offshore test are shown in Figure 21 below:

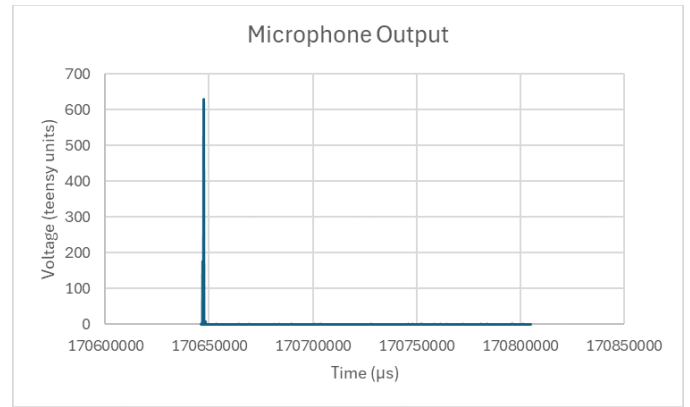


Figure 21. Microphone output from offshore test

The results show one large spike and no echoes after that. This made it impossible to guess a depth measurement from this test. It is possible that the ocean was too deep in this region for the microphone to hear an echo, however, the expected depth for the harbor was 9-15ft [6], a range that worked during testing in Axelrood Pool's deep end. The more likely explanation of this result is the complexity of the surface of the ocean. The pool's floor was flat meaning the sound waves bounce straight back to the surface. However, the ocean floor may have been variable with rocks and corals on the floor. These would cause the sound waves to bounce in multiple directions and not straight up back into the microphone.

VI. CONCLUSION

The AUV was able to measure salinity, temperature and ocean depth. The salinity was measured to be $33.4 \pm 0.2\text{ppt}$ for both nearshore and offshore tests. These measurements agreed with the range of expected ocean salinities [7].

The temperature was measured to decrease from 21.25 to $20.6 \pm 0.22^\circ\text{C}$ as the robot moved farther away from the shore.

These measurements were higher than the measured ground truth measurement of 18.7°C measured near the pier. The discrepancies were explained due to different measuring locations and calibration inconsistencies.

Depth was measured to be 3.93 ± 0.7 ft near the shore at Baby Beach. The measurement was supported by estimates of the water height based on the water reaching waist high. The depth was not able to be measured at deeper locations in the harbor. It was hypothesized that this was a result of complex ocean floor patterns causing sound waves to bounce in multiple directions.

The biggest takeaway from this project was learning how to implement and design sensors from scratch and making them readable. The AUV has the ability to collect useful data relating to impacts of climate change and human-caused factors. Another takeaway was related to limitations of sonar in complex environments. The Sonar sensor was reasonably accurate in a flat-bottom pool but struggled in the Dana Point harbor.

To improve this AUV the following changes are recommended:

- *GPS integration*: Implementing an accurate, reliable GPS tracing system to the AUV would allow the results to be visualized as a function of position rather than time allowing for a visual mapping of ocean conditions.
- *Thermistor calibration*: Keeping the calibration thermistor and data collection thermistor consistent could improve the accuracy of the results.
- *Salinity corrosion*: Significant corrosion buildup caused by salt water usage on the NEMA 1-15p connector was noticed and the probe had to be switched out to allow continuation of data collection. Switching probes could cause inconsistent measurements, so designing a probe that doesn't corrode could improve the accuracy of measurements
- *Sonar accuracy improvements*: At deeper ranges and complex environments, the sonar was unreliable. Future iterations could implement signal processing such as fast Fourier transforms allowing users to better distinguish speaker echoes from background noise and multipath interference

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